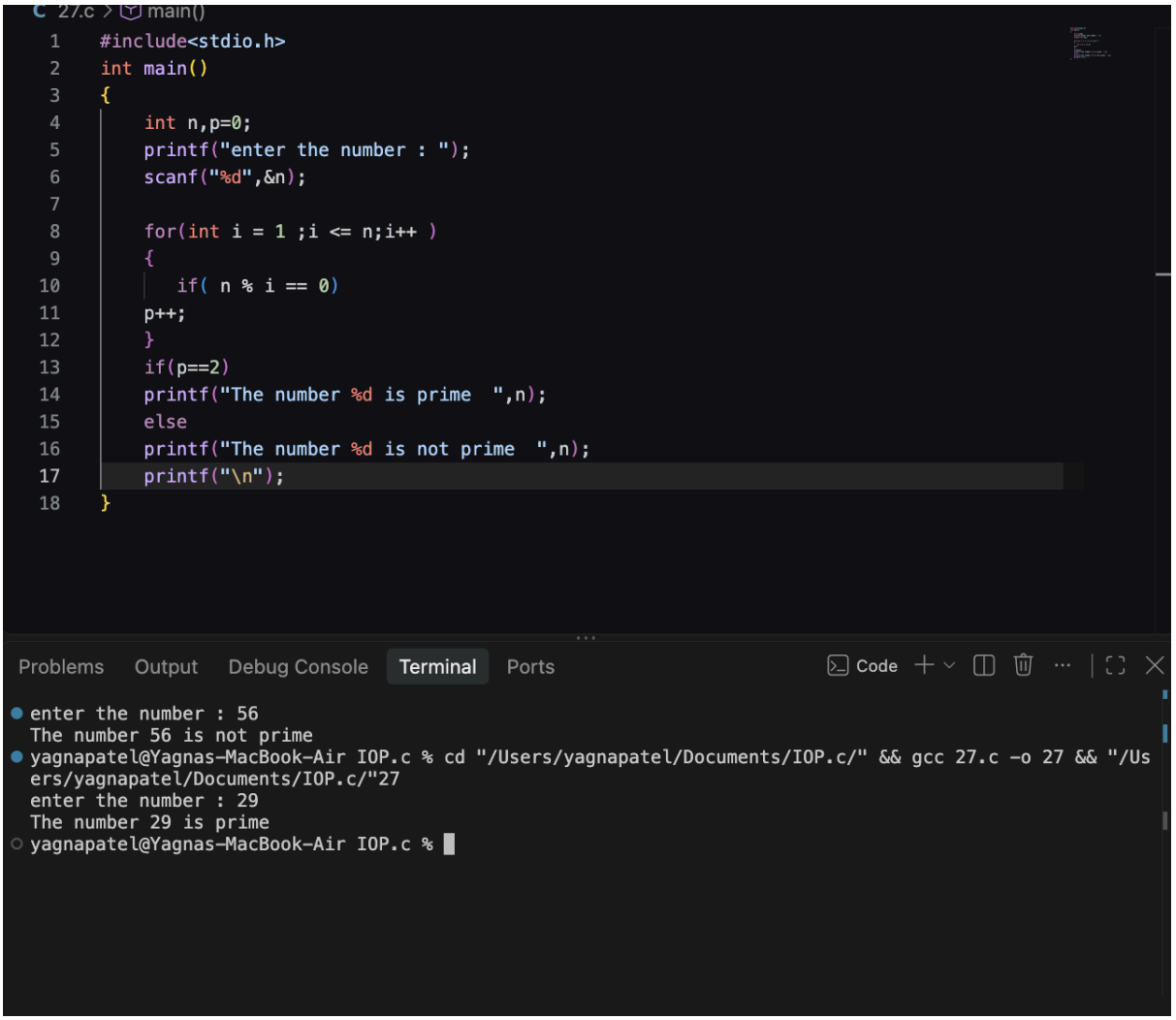
27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.

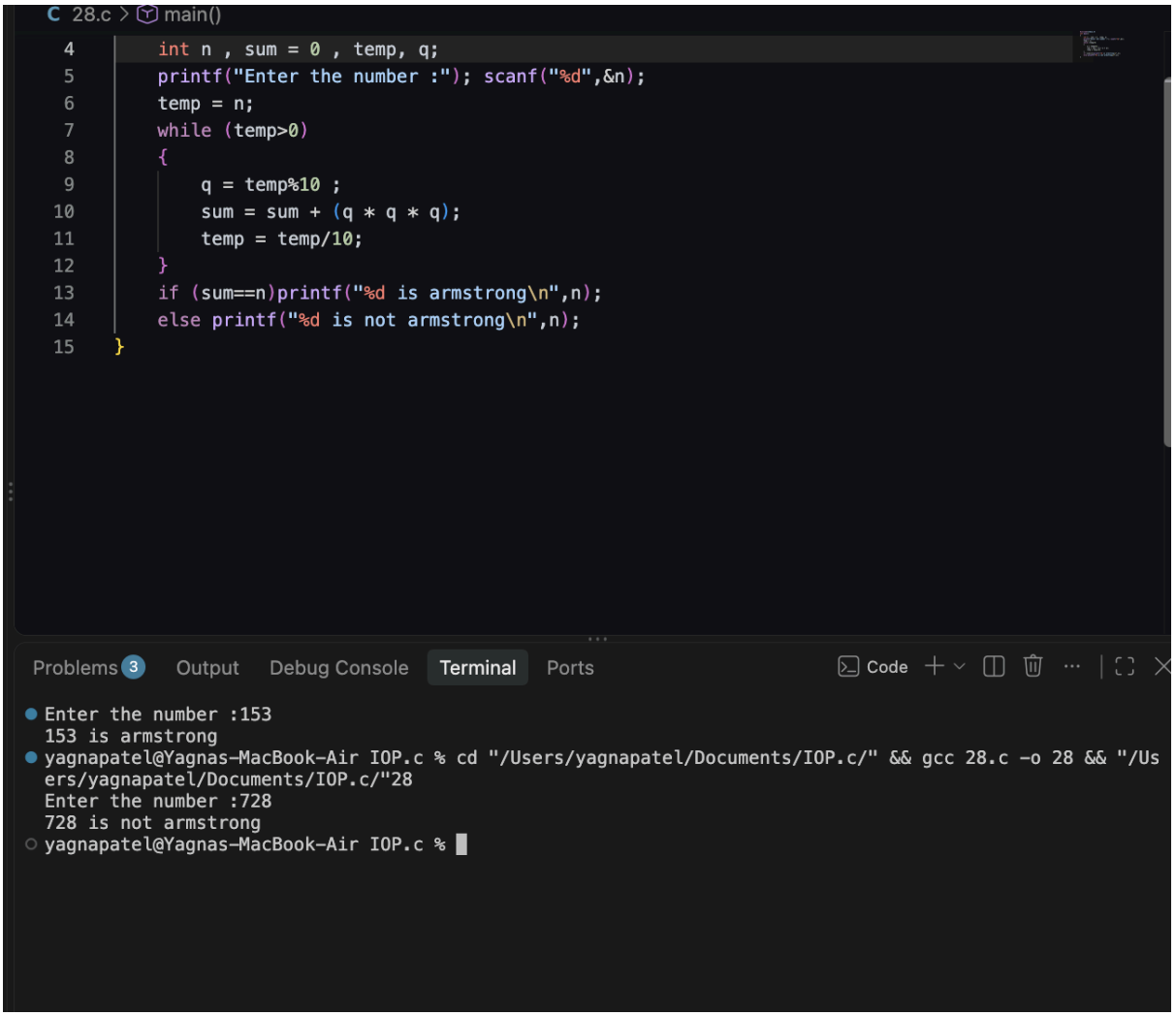


```c
#include<stdio.h>
int main()
{
    int n,p=0;
    printf("enter the number : ");
    scanf("%d",&n);

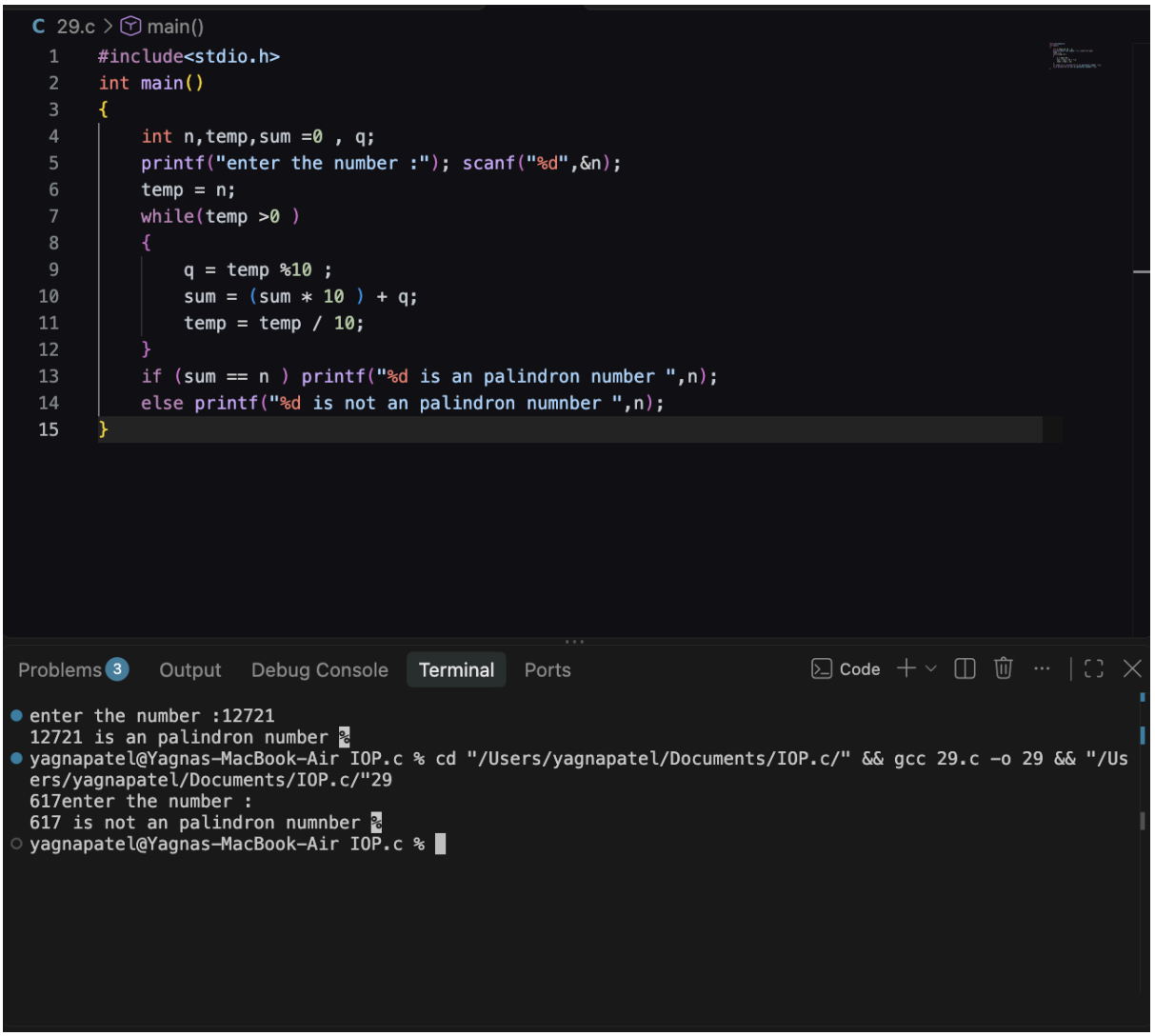
    for(int i = 1 ;i <= n;i++ )
    {
       if( n % i == 0)
    p++;
    }
    if(p==2)
    printf("The number %d is prime  ",n);
    else
    printf("The number %d is not prime  ",n);
    printf("\n");
}
```

```
enter the number : 56
The number 56 is not prime
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 27.c -o 27 && "/Users/yagnapatel/Documents/IOP.c/"27
enter the number : 29
The number 29 is prime
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

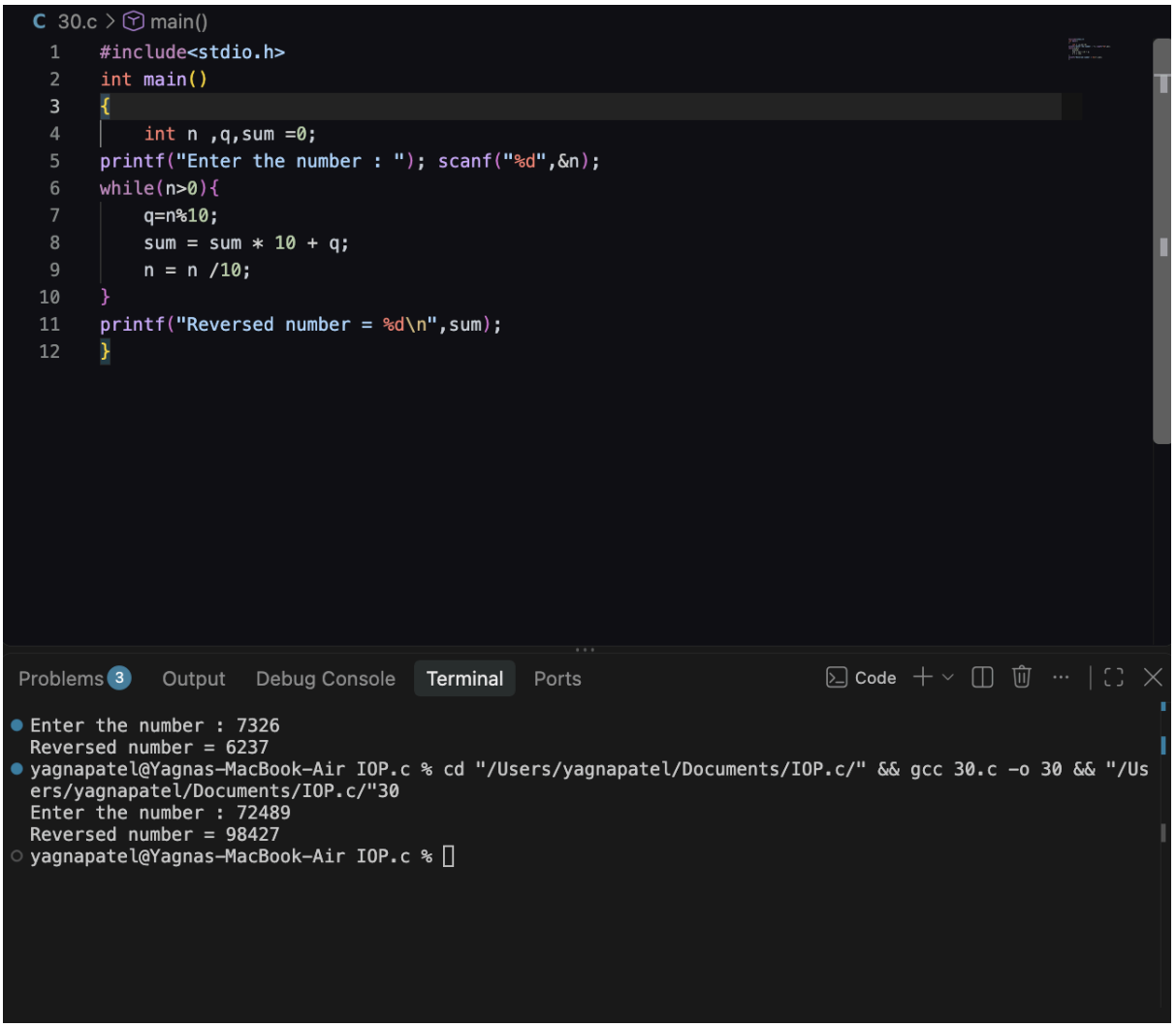
28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.



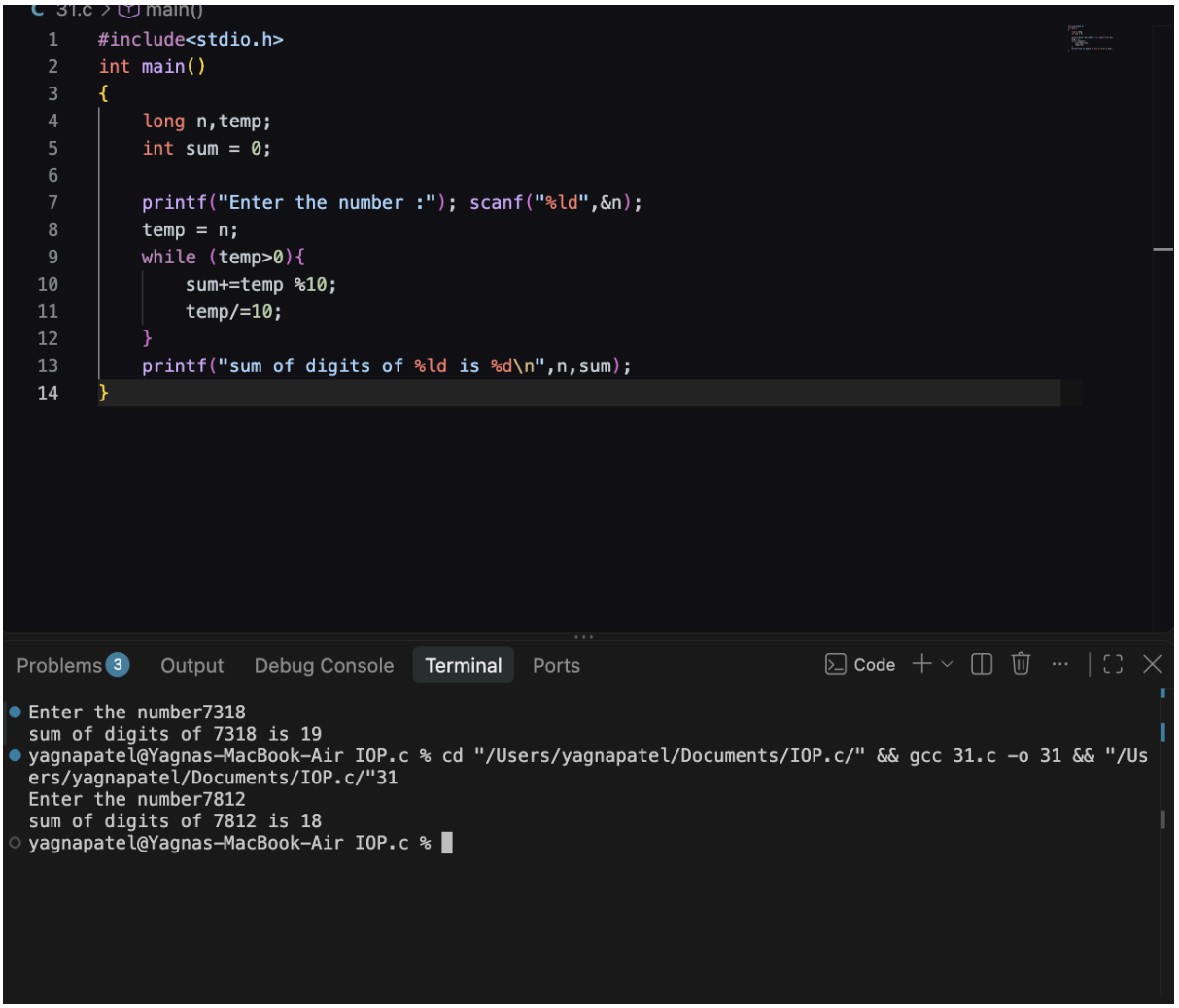
29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.



30. PROGRAM TO REVERSE OF A GIVEN NO



31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF A ‘N’ DIGIT NO.



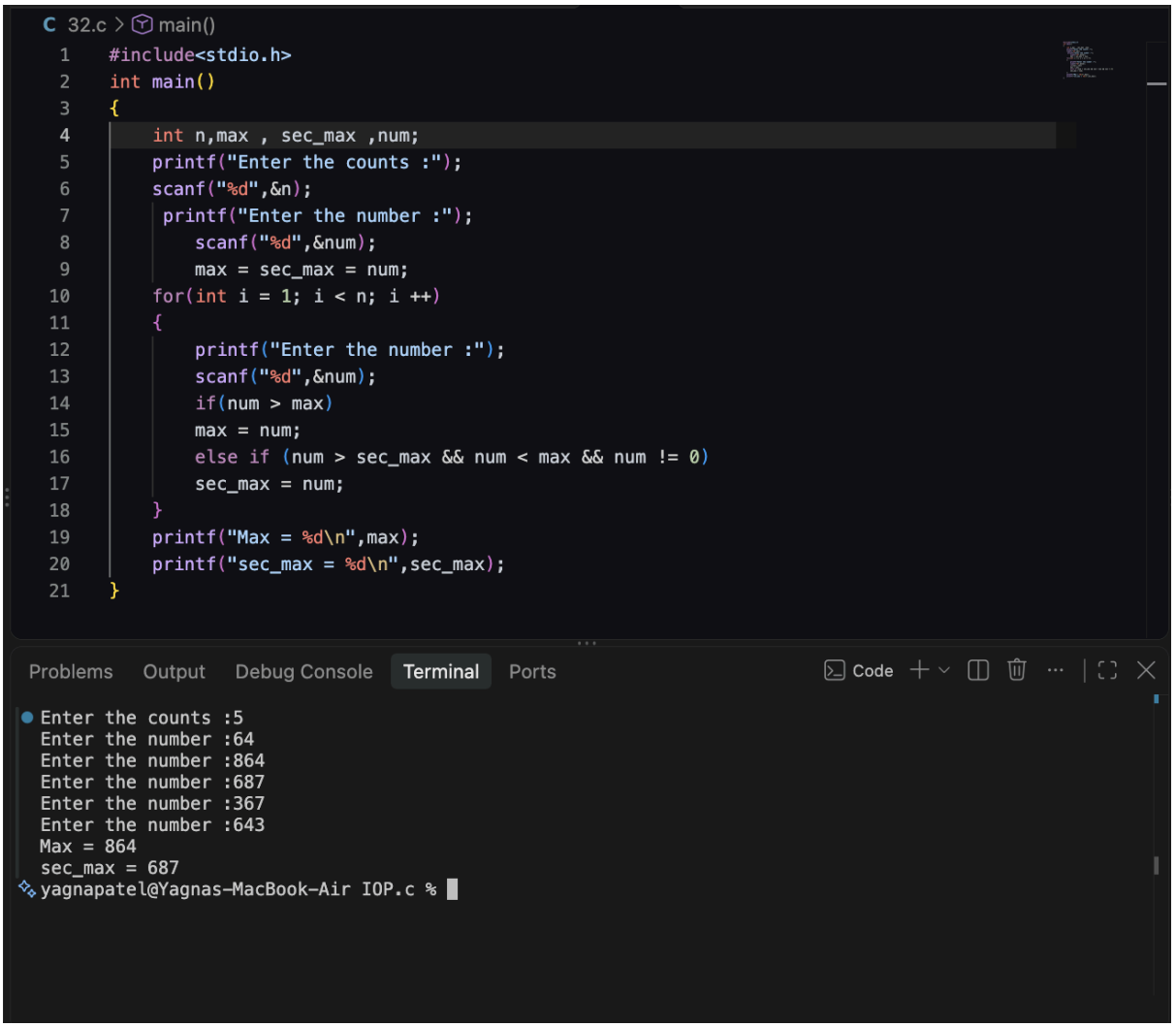
```c
#include<stdio.h>
int main()
{
    long n,temp;
    int sum = 0;

    printf("Enter the number :"); scanf("%ld",&n);
    temp = n;
    while (temp>0){
        sum+=temp %10;
        temp/=10;
    }
    printf("sum of digits of %ld is %d\n",n,sum);
}
```

```
Problems 3  Output  Debug Console  Terminal  Ports        Code

Enter the number7318
sum of digits of 7318 is 19
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 31.c -o 31 && "/Us
ers/yagnapatel/Documents/IOP.c/"31
Enter the number7812
sum of digits of 7812 is 18
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

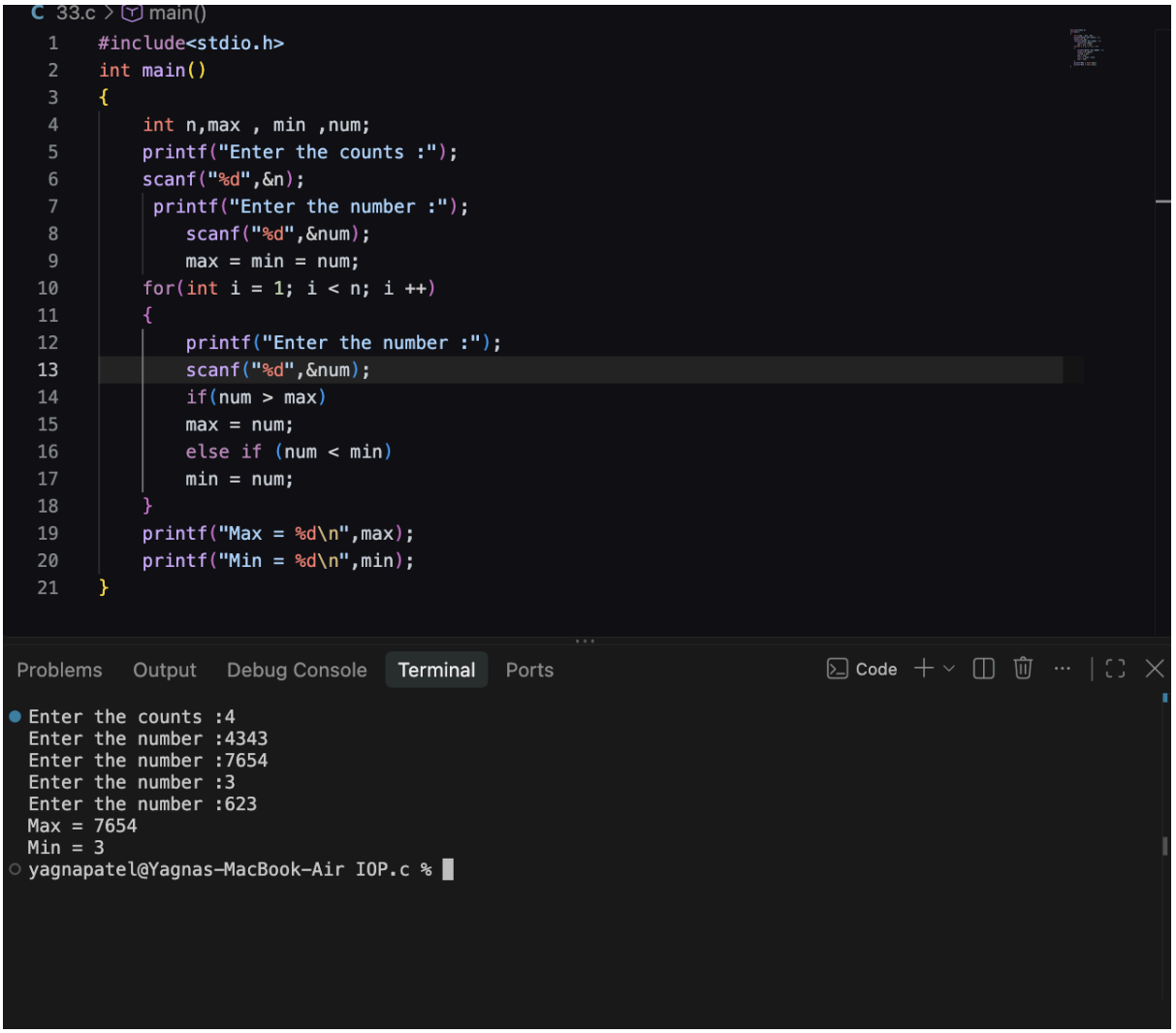
32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM ‘N’ NOS .



```c
#include<stdio.h>
int main()
{
    int n,max , sec_max ,num;
    printf("Enter the counts :");
    scanf("%d",&n);
     printf("Enter the number :");
        scanf("%d",&num);
        max = sec_max = num;
    for(int i = 1; i < n; i ++)
    {
        printf("Enter the number :");
        scanf("%d",&num);
        if(num > max)
        max = num;
        else if (num > sec_max && num < max && num != 0)
        sec_max = num;
    }
    printf("Max = %d\n",max);
    printf("sec_max = %d\n",sec_max);
}
```

```
Enter the counts :5
Enter the number :64
Enter the number :864
Enter the number :687
Enter the number :367
Enter the number :643
Max = 864
sec_max = 687
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

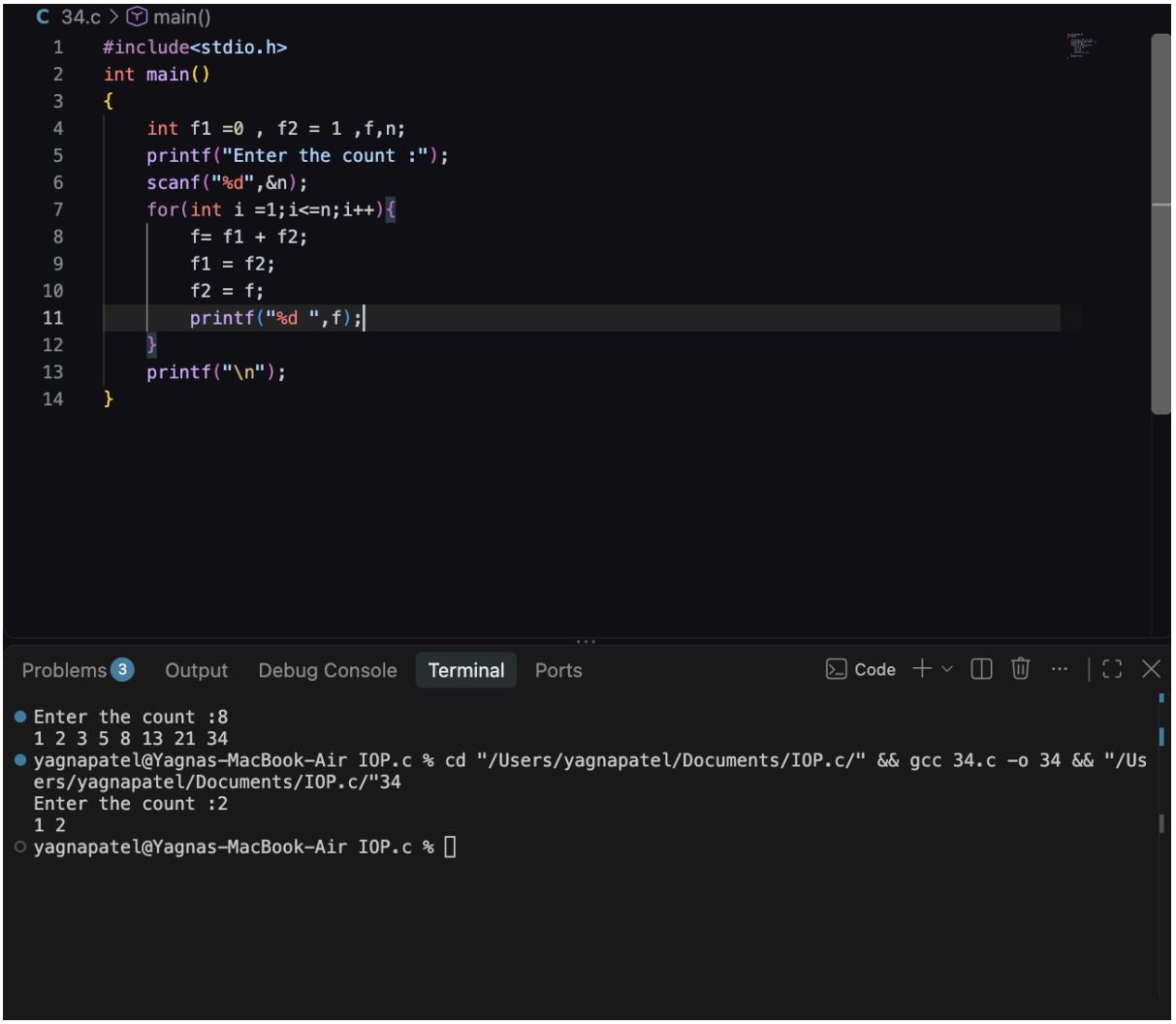
33. PROGRAM TO FINDOUT MAX AND MIN FROM ‘N’ NOS.



```c
#include<stdio.h>
int main()
{
    int n,max , min ,num;
    printf("Enter the counts :");
    scanf("%d",&n);
     printf("Enter the number :");
        scanf("%d",&num);
        max = min = num;
    for(int i = 1; i < n; i ++)
    {
        printf("Enter the number :");
        scanf("%d",&num);
        if(num > max)
        max = num;
        else if (num < min)
        min = num;
    }
    printf("Max = %d\n",max);
    printf("Min = %d\n",min);
}
```

```
Enter the counts :4
Enter the number :4343
Enter the number :7654
Enter the number :3
Enter the number :623
Max = 7654
Min = 3
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

34. PROGRAM TO PRINT FIBONACCI SERIES.(0,1,1,2,3…)



```c
#include<stdio.h>
int main()
{
    int f1 =0 , f2 = 1 ,f,n;
    printf("Enter the count :");
    scanf("%d",&n);
    for(int i =1;i<=n;i++){
        f= f1 + f2;
        f1 = f2;
        f2 = f;
        printf("%d ",f);
    }
    printf("\n");
}
```

```
Enter the count :8
1 2 3 5 8 13 21 34
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 34.c -o 34 && "/Users/yagnapatel/Documents/IOP.c/"34
Enter the count :2
1 2
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

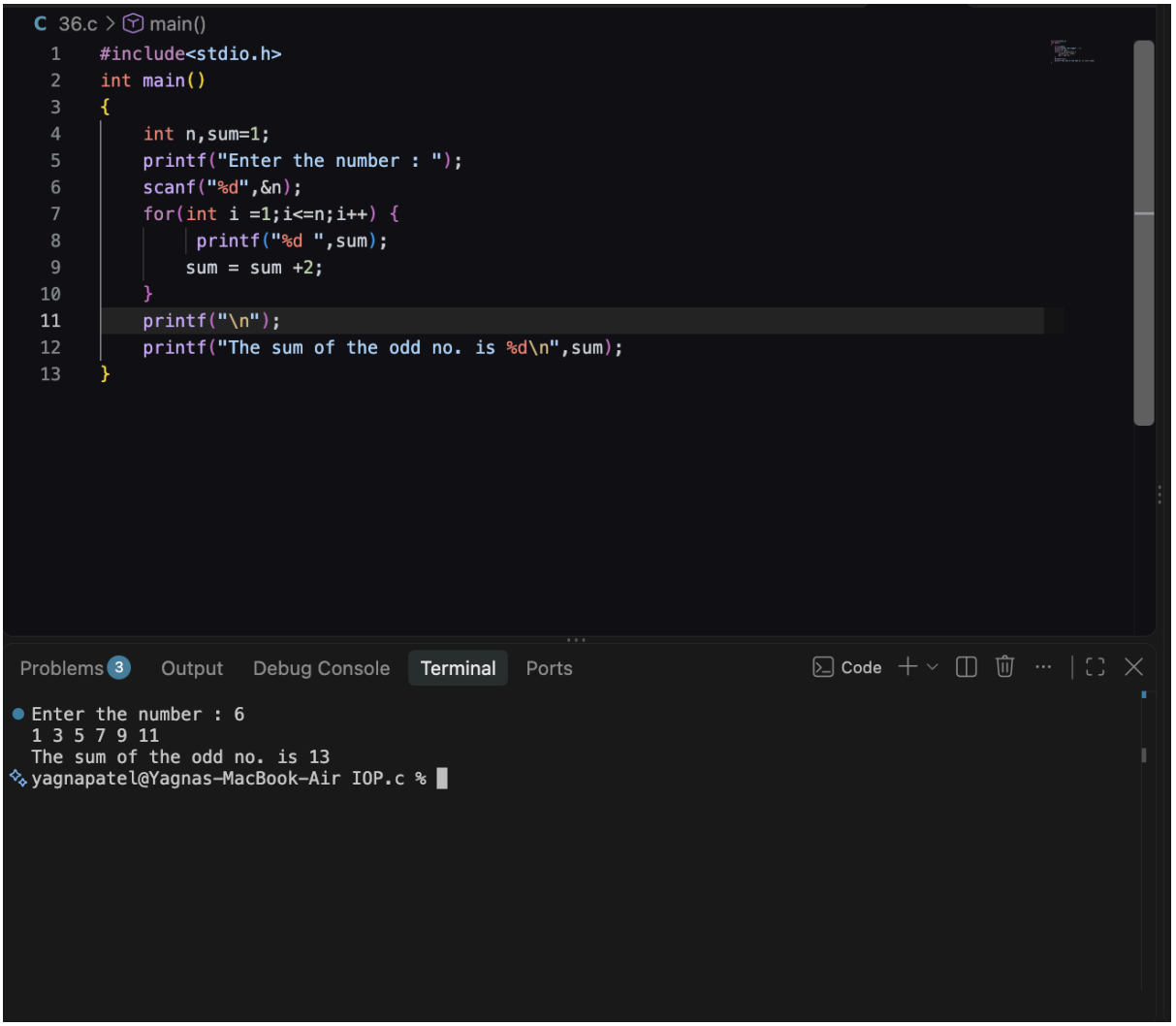
35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N

```c
#include<stdio.h>
int main()
{
    int n,sum =0 ,temp;
    printf("Enter the number :");
    scanf("%d",&n);
    while (n > 9){
        while(n > 0){
            sum+=n%10;
            n = n/10;
        }
        n = sum;
        sum =0;
    }
    printf("The sum is %d\n",n);
}
```

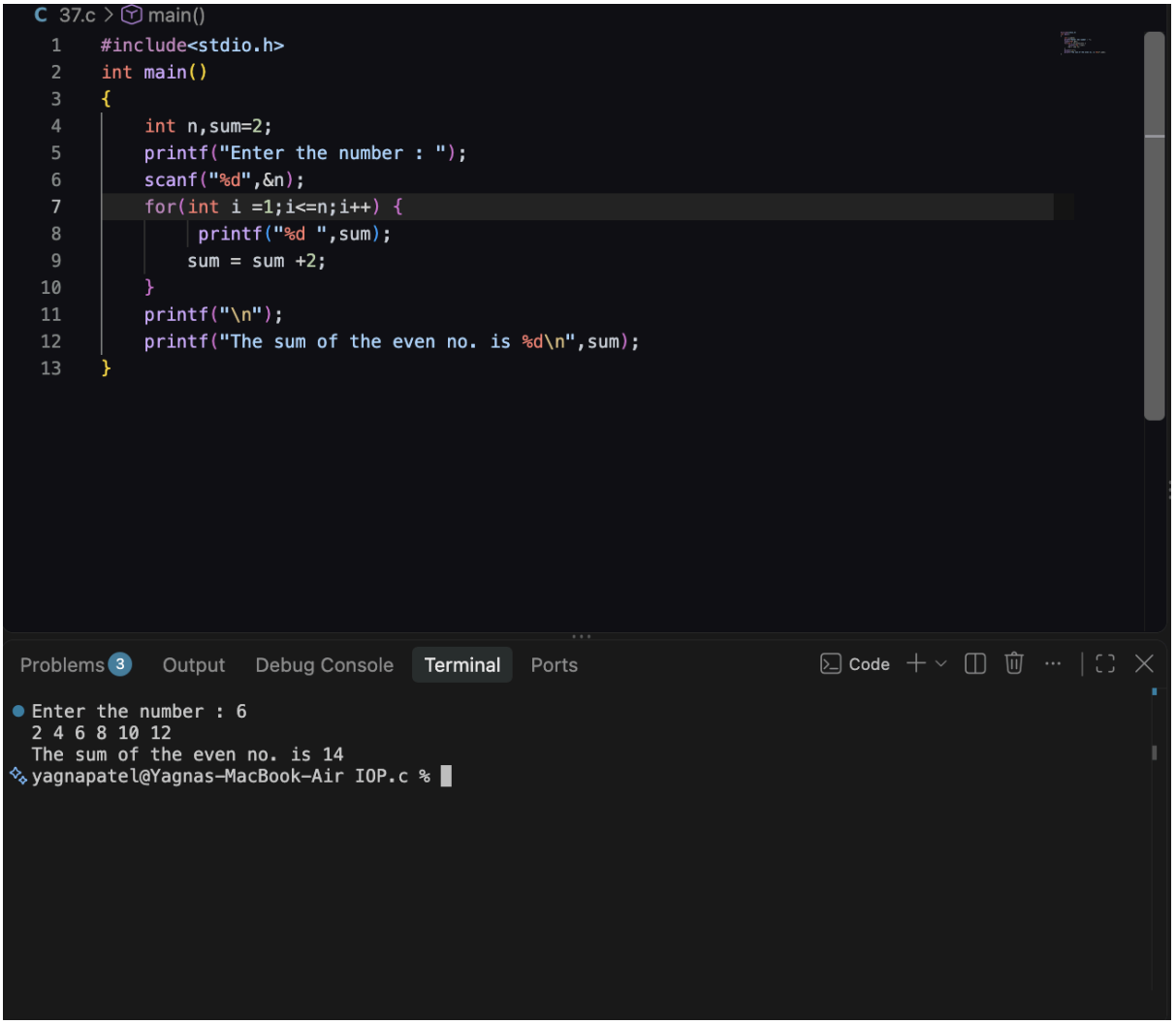
```
Problems 3  Output  Debug Console  Terminal  Ports

Enter the number :561
The sum is 3
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 35.c -o 35 && "/Users/yagnapatel/Documents/IOP.c/"35
Enter the number :618
The sum is 6
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

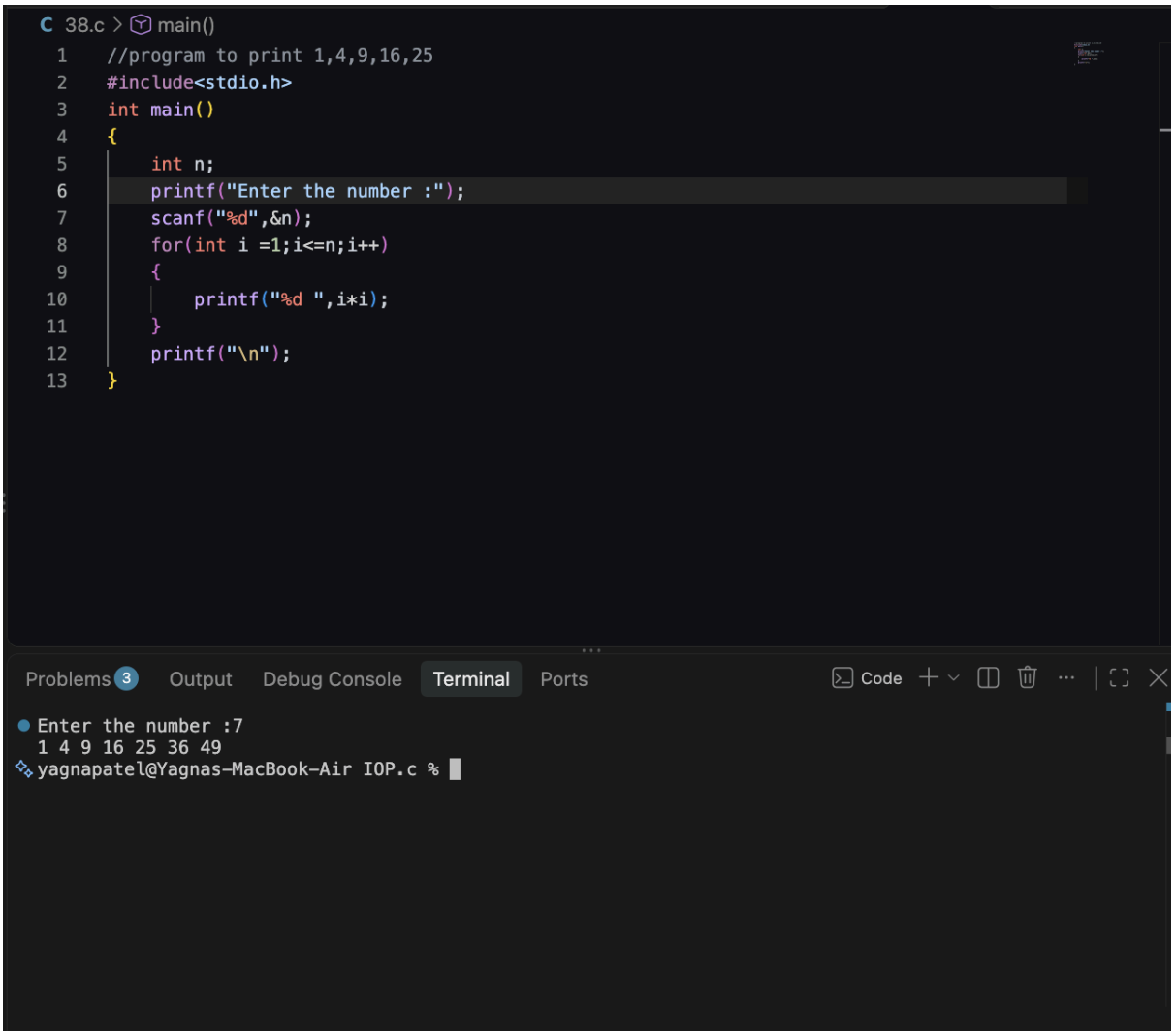
36. PROGRAM TO PRINT 1,3,5,7,9……N.



37. PROGRAM TO PRINT 2,4,6,8,10,12……N.



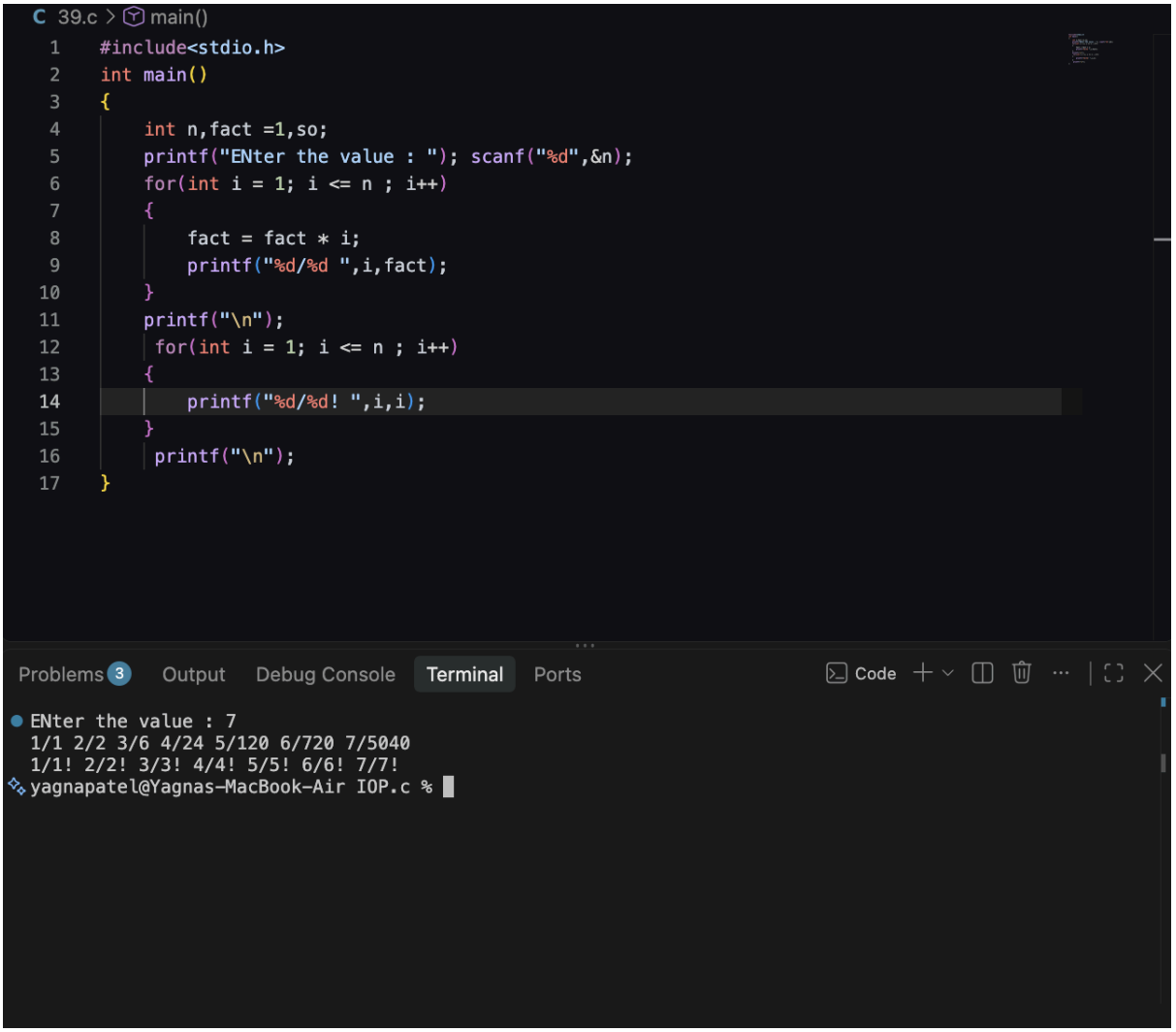
38. PROGRAM TO PRINT 1,4,9,16,25,……N.



```c
//program to print 1,4,9,16,25
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the number :");
    scanf("%d",&n);
    for(int i =1;i<=n;i++)
    {
        printf("%d ",i*i);
    }
    printf("\n");
}
```

```
Enter the number :7
1 4 9 16 25 36 49
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

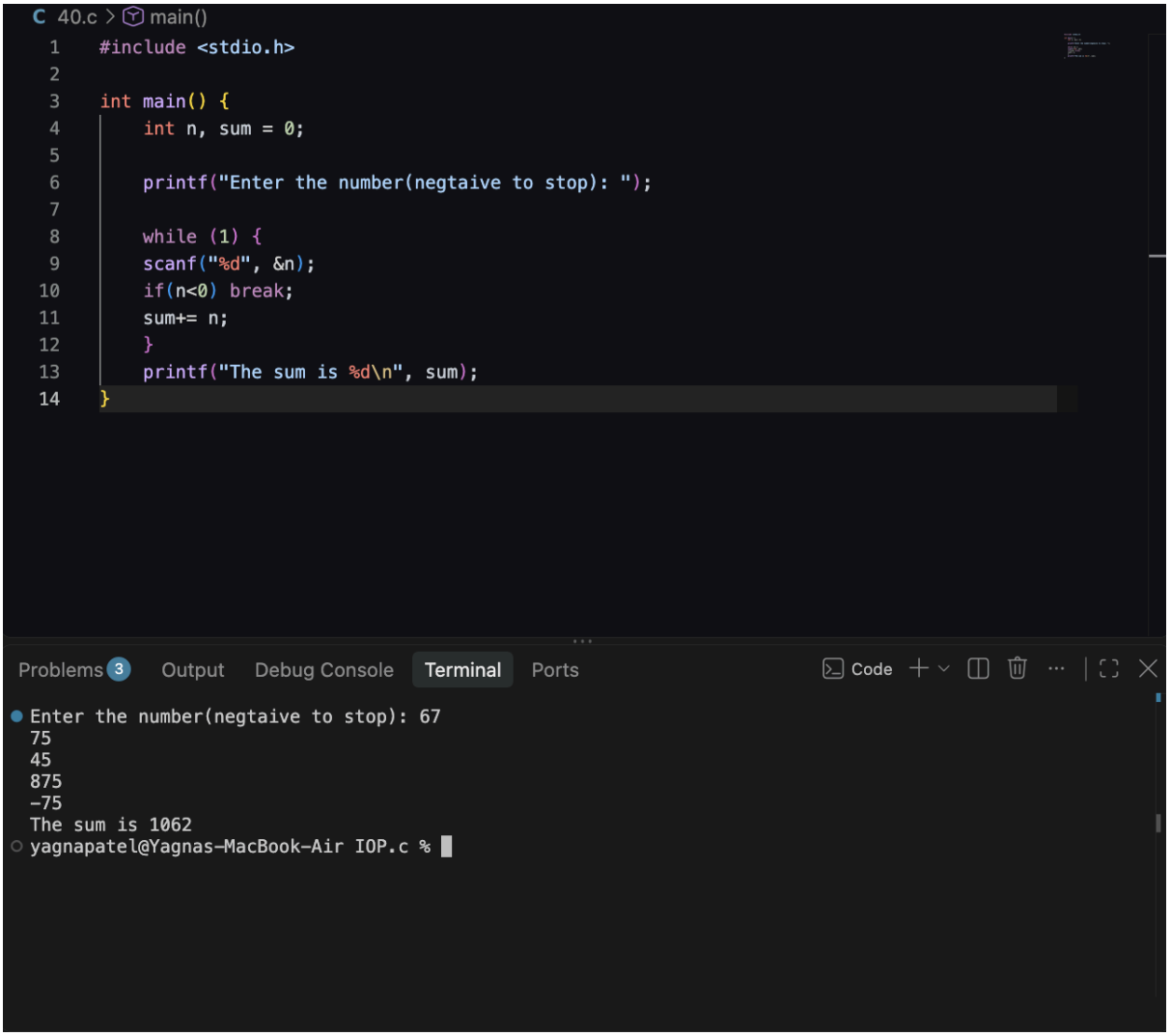
39. PROGRAM TO PRINT 1/1!+2/2!+3/3!+4/4!,……N.



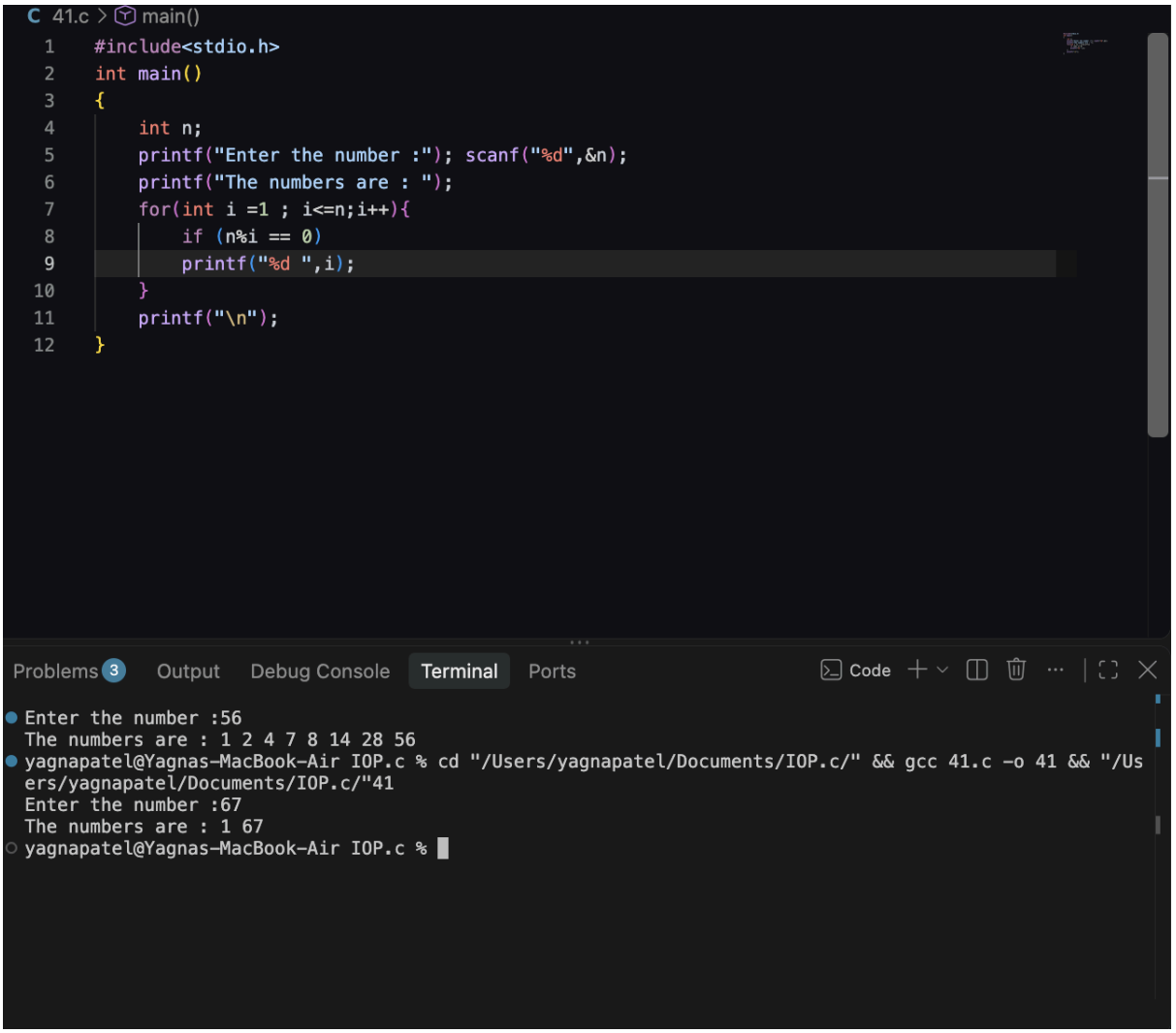
```c
#include<stdio.h>
int main()
{
    int n,fact =1,so;
    printf("ENter the value : "); scanf("%d",&n);
    for(int i = 1; i <= n ; i++)
    {
        fact = fact * i;
        printf("%d/%d ",i,fact);
    }
    printf("\n");
     for(int i = 1; i <= n ; i++)
    {
        printf("%d/%d! ",i,i);
    }
     printf("\n");
}
```

```
ENter the value : 7
1/1 2/2 3/6 4/24 5/120 6/720 7/5040
1/1! 2/2! 3/3! 4/4! 5/5! 6/6! 7/7!
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

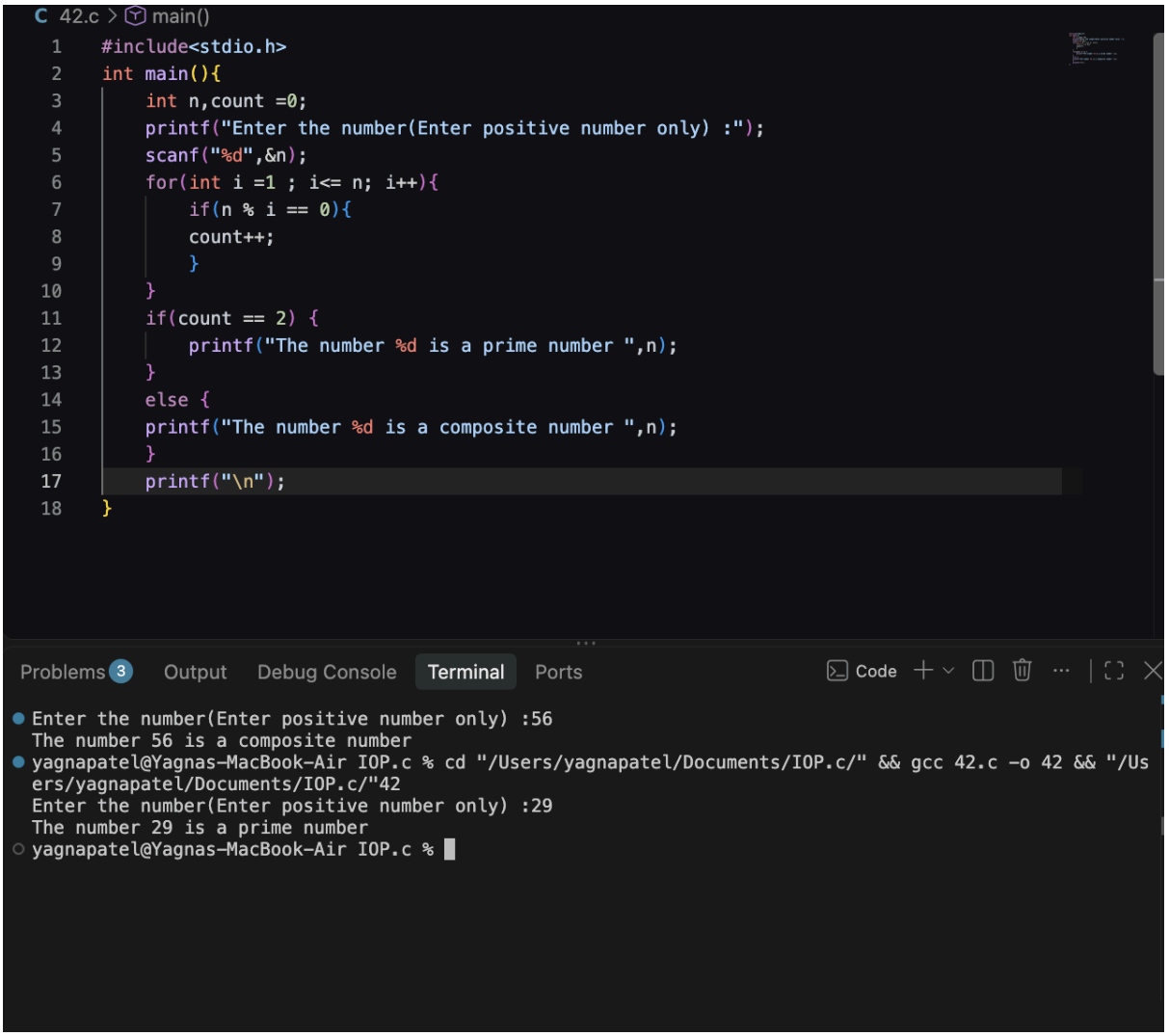
40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.



41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)



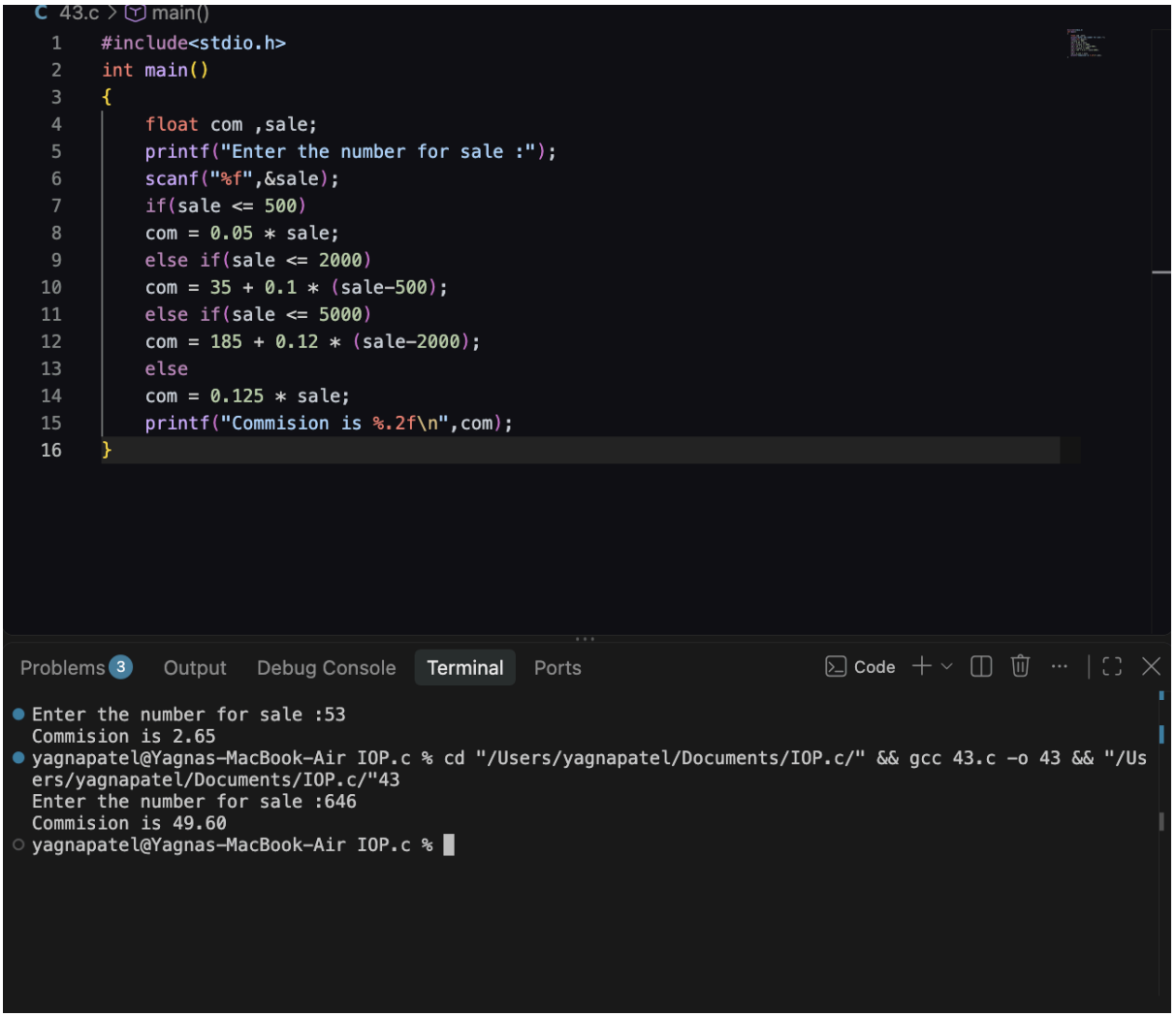
42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.



```c
#include<stdio.h>
int main(){
    int n,count =0;
    printf("Enter the number(Enter positive number only) :");
    scanf("%d",&n);
    for(int i =1 ; i<= n; i++){
        if(n % i == 0){
        count++;
        }
    }
    if(count == 2) {
        printf("The number %d is a prime number ",n);
    }
    else {
    printf("The number %d is a composite number ",n);
    }
    printf("\n");
}
```

```
Enter the number(Enter positive number only) :56
The number 56 is a composite number
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 42.c -o 42 && "/Users/yagnapatel/Documents/IOP.c/"42
Enter the number(Enter positive number only) :29
The number 29 is a prime number
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES
AMOUNT GIVEN BELOW.
If If If sales<=Rs. 500, commission is 5%
sales> but <=2000, commission is Rs. 35 plus 10% above Rs. 500
sales>2000 but <=5000, commission is Rs. 185 plus 12% above Rs. 2000
If sales>5000, commission is 12.5%



44. WRITE A PROGRAM IN C FOR THE FOLLOWING.
AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS

